Market Basket

Mini Project Presentation





Market Basket Analysis

Muhammad Alfi Syahputra

2026

Introduction



SUMMARY

Graduated from President University majoring in International Relations with a broad understanding of global affairs, Indonesian politics, gender issues, and popular culture. A self-motivated self-starter that has improved his expertise and grown professionally. Highly interested in Data Analyst, with skills in data processing, building data pipelines, and big data analysis. Proficient in Python, SQL, and tools such as Tableau and PowerBI. Known for being detail-oriented, adaptable, and capable of working both independently and collaboratively. Ready to leverage prior work experience to support data management and optimization, contributing to data-driven decision-making.

WORK EXPERIENCES

Dinas Kesehatan Kota Bandung **July - November 2021**

Public Relations Staff

- Documenting and promoting an event created for the distribution of the Covid-19 vaccines
- Managed and organized participant data to ensure accuracy, consistency, and easy retrieval.
- Utilized Microsoft Excel or Google Sheets to organize, update, and manage data systematically.

EDUCATION

Dibimbing Bootcamp as Data Analyst **2026 - Present**

- Studied data storytelling, visualization, and performance measurement using industry tools.
- Technical Skills: Excel (pivot tables, data cleaning), SQL, Tableau, Power BI, Google Collab.
- Capstone Project: Designed an interactive dashboard to analyze company performance metrics and KPIs.

President University / International Relations / GPA (3.49 / 4.00) **2018 - 2022**

Project

Project Details



Objective

To compare sales performance across regions and identify the main contributors to sales and profit.

Purpose

My International Relations background taught me to understand regional differences and analyze issues from different perspectives. I applied this knowledge to compare sales across regions and understand how product categories and customer segments contribute to business performance.

Outcome

- The West region had the highest sales, followed by the East region.
- Technology was the top-selling product category, while the Consumer segment contributed the most sales.
- The analysis showed that higher sales do not always mean higher profit.
- This project helped me apply regional analysis skills to business data and develop data-driven recommendations.

Repository Superstore Analysis (Regional Sales Comparison)

Project Details



Objective

To analyze marketing channel performance, customer behavior, and conversion rates to identify opportunities to improve marketing results.

Purpose

My International Relations studies helped me develop communication and analytical skills. My previous experience in Public Relations also taught me the importance of understanding audiences. These experiences motivated me to analyze customer behavior and understand how marketing channels influence purchasing decisions.

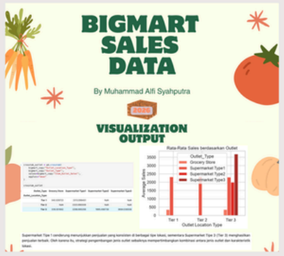
Outcome

- Paid Ads generated the highest number of purchases, followed by Social Media and Organic Search.
- Instagram showed the largest drop-off between impressions and clicks.
- The analysis identified opportunities to improve advertising, customer conversion, and repeat purchases.
- This project helped me combine my communication background with data analysis to support marketing decisions.

Repository Marketing Growth (Customer Journey)



Project Details



Objective

To analyze BigMart sales data and identify factors related to product sales, pricing, and outlet performance using Python.

Purpose

As an International Relations graduate, I learned to think critically, analyze problems, and understand relationships between different factors. I applied these skills to sales data to understand business performance and identify opportunities for improvement.

Outcome

- Tier 2 outlets showed strong average sales, while Tier 3 outlets contributed significantly to overall sales.
- The analysis identified opportunities to improve product promotions and inventory management.
- This project helped me apply my analytical thinking to business problems and develop my skills as a Data Analyst.

Repository Bigmart Sales Data

Project Details



Objective

To analyze Bigmart sales data to understand how product price, outlet location, and product characteristics affect sales performance.

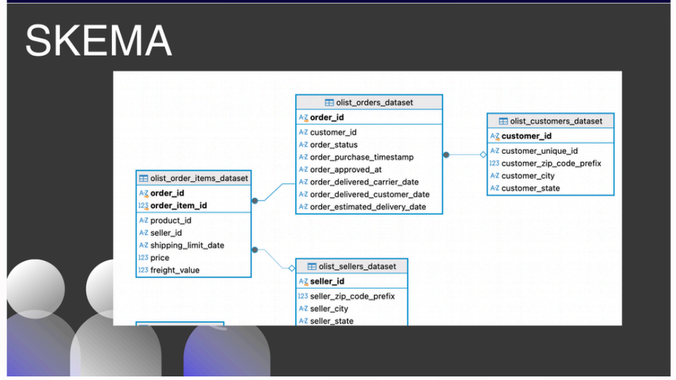
Purpose

My International Relations background taught me to compare regions and look at one issue from several angles. I applied the same approach to compare sales across outlet location tiers and product types, so I could see which areas and products drive sales and where opportunities are being missed. Moving into data analytics, I used Python to clean the data (missing values, duplicates, outliers) so the analysis rests on reliable data.

Outcome

- Tier 2 outlets had the highest average sales, followed by Tier 3, while Tier 1 was the lowest.
- The Household category has an average MRP similar to other categories but lower sales than top categories such as Starchy Foods, which suggests room to improve marketing and pricing.
- Fat content (Regular vs Low Fat) showed no consistent pattern in sales, so item type matters more than fat content.
- Recommendations: run bundling or discounts for Household products, prioritize stock for high-sales outlets (Tier 2), and focus promotion on item types with high demand.

Repository Bigmart Sales Data (2)



Journal



AIP Publishing

AIP Conference Proceedings

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Volume 2772, Issue 1

24 February 2023

The 3rd Faculty of Industrial Technology International Congress 2021 International Conference

Enriching Engineering Science through Collaboration of Multidisciplinary Fields

Bandung, Indonesia • 28–29 October 2021

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RESEARCH ARTICLE | FEBRUARY 24 2023

Determining combination of items for product bundle offering using market basket analysis

Hendang Setyo Rukmi ; Fahmi Arif; Josua

+ Author & Article Information

AIP Conf. Proc. 2772, 080005 (2023)

https://doi.org/10.1063/5.0115308

Determining the combination of items for product bundle offerings is important to increase consumer buying interest. By analyzing historical data on sales transactions, purchasing patterns can be identified to find the right product combination that has a high probability of selling. This study aims to determine consumer purchasing patterns on large transaction data using market basket analysis

https://pubs.aip.org/aip/acp/article-abstract/2772/1/080005/2873223/Determining-combination-of-items-for-product?redirectedFrom=fulltext

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PROBLEM STATEMENT

DATA UNDERSTANDING

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Market Basket

Problem Statement



Bagaimana perkembangan perilaku belanja pelanggan, dan produk apa yang menjadi penggerak utamanya?

Apakah produk yang paling laris juga saling berkaitan, dan pasangan produk mana yang benar-benar memiliki hubungan pembelian yang kuat

Strategi apa yang sebaiknya diambil toko untuk memanfaatkan pola belanja tersebut?

Market Basket

Data Understanding

	Member_number	Date	itemDescription
0	1808	21-07-2015	tropical fruit
1	2552	05-01-2015	whole milk
2	2300	19-09-2015	pip fruit
3	1187	12-12-2015	other vegetables
4	3037	01-02-2015	whole milk

```
df = pd.read_csv("Market Basket Analysis- Groceries_dataset.csv")
df
```

Dataset memiliki dimensi :

Jumlah Baris = 38765

Jumlah Kolom = 3

Sumber data:
https://drive.google.com/drive/folders/1knQlrdq0dpJ8o9M9faVt8n3ZlMKRY_vP

Market Basket

Column Description

Kolom	Tipe	Deskripsi	Contoh
`Member_number`	Angka (ID)	Nomor ID pelanggan/member yang berbelanja. Ada 3.898 member unik.	1808
`Date`	Teks (format dd-mm-yyyy)	Tanggal pembelian, rentang 1 Jan 2014 sampai 30 Des 2015. Di Tableau perlu diubah ke tipe tanggal.	21-07-2015
`itemDescription`	Teks	Nama produk yang dibeli. Ada 167 produk unik. Satu baris = satu produk dalam satu pembelian.	tropical fruit

Market Basket

Data Type

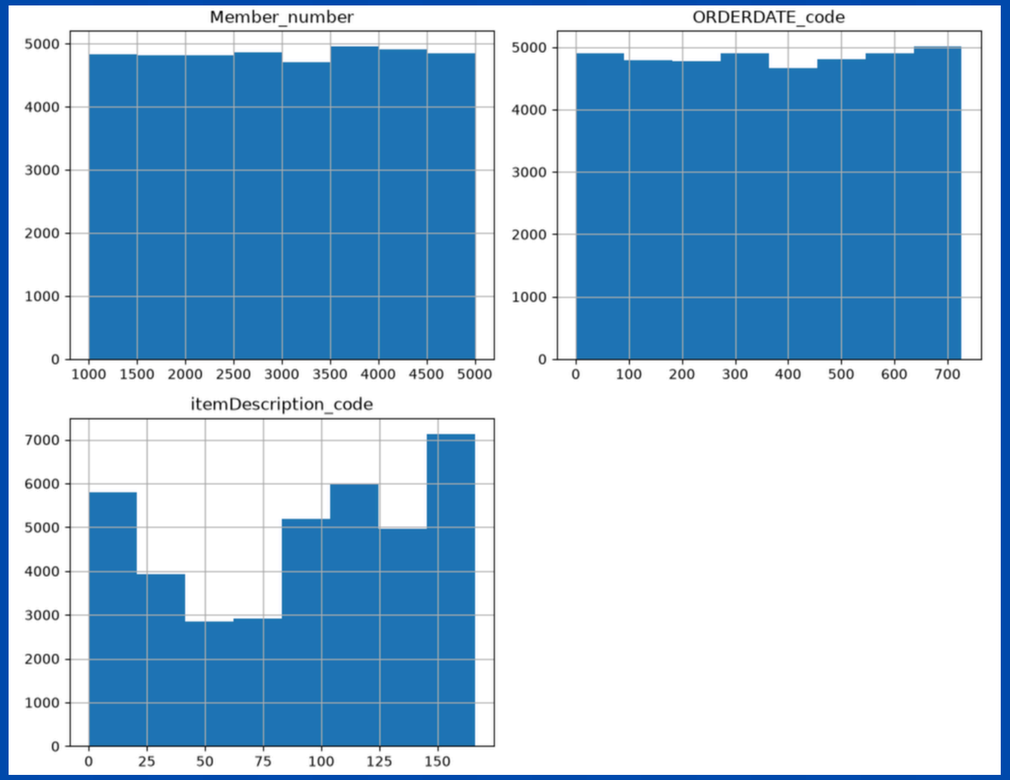
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 38765 entries, 0 to 38764
Data columns (total 3 columns):
 #   Column           Non-Null Count  Dtype 
---  ------           --------------  ----- 
 0   Member_number    38765 non-null  int64 
 1   Date             38765 non-null  str   
 2   itemDescription  38765 non-null  str   
dtypes: int64(1), str(2)
memory usage: 1.6 MB
```

Tipe data tidak sesua pada kolom Date, yang nanti nya akan diubah menjadi datetime

Market Basket

Data Distribution



Distribusi setelah melakukan Encoder

Market Basket

Outlier Checking

Outlier

```
cek_outlier = df_copy.select_dtypes(include=["int64"]).columns
cek_outlier
```

```
Index(['Member_number', 'ORDERDATE_code', 'itemDescription_code'], dtype='str')
```

```
for i in cek_outlier :
    Q1 = df_copy[i].quantile(0.25)
    Q3 = df_copy[i].quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    outlier = (df_copy[i] < lower) | (df_copy[i] > upper)
    jumlah_outlier = outlier.sum() # Mengetahui jumlah outlier pada kolom
    persentase_outlier = jumlah_outlier/len(outlier) * 100 # Mengetahui persentase outlier pada kolom
    print(f"Jumlah Outlier pada Kolom {i} : {jumlah_outlier}")
    print(f"Persentase Outlier pada Kolom {i} : {persentase_outlier:.2f}%\n")
```

```
Jumlah Outlier pada Kolom Member_number : 0
Persentase Outlier pada Kolom Member_number : 0.00%

Jumlah Outlier pada Kolom ORDERDATE_code : 0
Persentase Outlier pada Kolom ORDERDATE_code : 0.00%

Jumlah Outlier pada Kolom itemDescription_code : 0
Persentase Outlier pada Kolom itemDescription_code : 0.00%
```

Outlier Checking setelah melakukan Encoder

Market Basket

Cek Konsistensi

CEK KONSISTENSI

```
konsistensi = df_copy.select_dtypes(include=["str"]).columns
```

```
for i in konsistensi :
    print(f"Kolom {i} :")
    print(df_copy[i].unique())
    print("\n")
```

```
Kolom Date :
<ArrowStringArray>
['21-07-2015', '05-01-2015', '19-09-2015', '12-12-2015', '01-02-2015',
 '14-02-2015', '08-05-2015', '23-12-2015', '20-03-2015', '12-02-2015',
 ...
 '28-06-2014', '18-08-2014', '25-11-2014', '03-11-2014', '19-07-2014',
 '11-02-2014', '15-09-2014', '10-02-2014', '29-06-2014', '12-11-2014']
Length: 728, dtype: str


Kolom itemDescription :
<ArrowStringArray>
[       'tropical fruit',            'whole milk',             'pip fruit',
      'other vegetables',            'rolls/buns',           'pot plants',
          'citrus fruit',                  'beef',         'frankfurter',
               'chicken',
 ...
         'flower (seeds)',                 'rice',                   'tea',
         'salad dressing',  'specialty vegetables',       'pudding powder',
           'ready soups',       'make up remover',       'toilet cleaner',
 'preservation products']
Length: 167, dtype: str
```

Untuk kolom semua penulisannya sudah konsisten

Market Basket

Duplicate & Missing Value

Missing Value

```
miss_value = df_copy.isnull().sum() # Mencari jumlah missing value
miss_percentage = miss_value / len(df_copy) * 100 # Mencari persentase missing value
```

```
print(f"Jumlah missing value : \n{miss_value}")
print(f"Persentase missing value : \n{miss_percentage}")
```

```
Jumlah missing value :
Member_number      0
Date               0
itemDescription    0
dtype: int64
Persentase missing value :
Member_number      0.0
Date               0.0
itemDescription    0.0
dtype: float64
```

Duplicate

```
df_copy.duplicated().sum()
```

```
np.int64(759)
```

```
duplicate_persentase = df_copy.duplicated().mean() * 100

print(f"Persentase duplicate: {duplicate_persentase:.2f}%")
```

```
Persentase duplicate: 1.96%
```

Early Insight

Insight dari data exploration yang bisa disimpulkan yaitu, dataset memiliki variabel numerik dan kategorikal, tidak ditemukannya missing value, pada kolom Date perlu dibuah tipe datanya supaya menjadi format datetime agar lebih sesuai dan bukan string.Tidak terdapat outlier. Terdapat ducplicate yang nantinya akan ditangani

Data Preparation

Penanganan datetime

```
df_copy["Date"] = pd.to_datetime(
    df_copy["Date"],
    format="%d-%m-%Y",
    errors="coerce")
```

```
df_copy.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 38765 entries, 0 to 38764
Data columns (total 5 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Member_number         38765 non-null  int64
 1   Date                  38765 non-null  datetime64[us]
 2   itemDescription       38765 non-null  str
 3   ORDERDATE_code        38765 non-null  int64
 4   itemDescription_code  38765 non-null  int64
dtypes: datetime64[us](1), int64(3), str(1)
memory usage: 1.9 MB
```

Penangan Duplicate

```
df_copy = df_copy.drop_duplicates()
df_copy.duplicated().sum()
```

```
np.int64(0)
```

Output Visualization

Year of Date
- 2014
- 2015

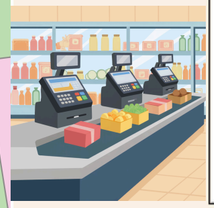
Itemsets
- bottled water
- citrus fruit
- other vegetables
- rolls/buns
- root vegetables
- sausage
- soda
- tropical fruit
- whole milk
- yogurt

Lift
0,58 — 1,23

Kategori Lift
- Negatif (<1)
- Positif (>1)

Lift
- 0,58
- 0,80
- 1,00
- 1,20
- 1,45

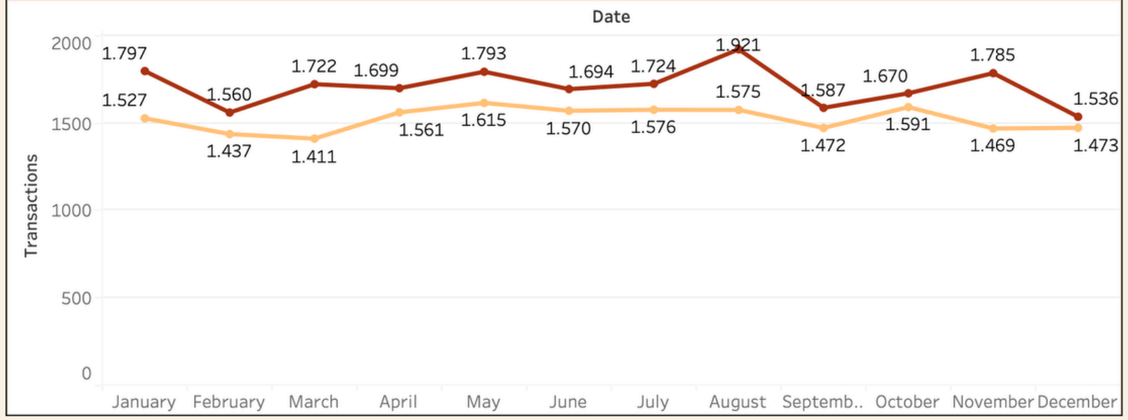




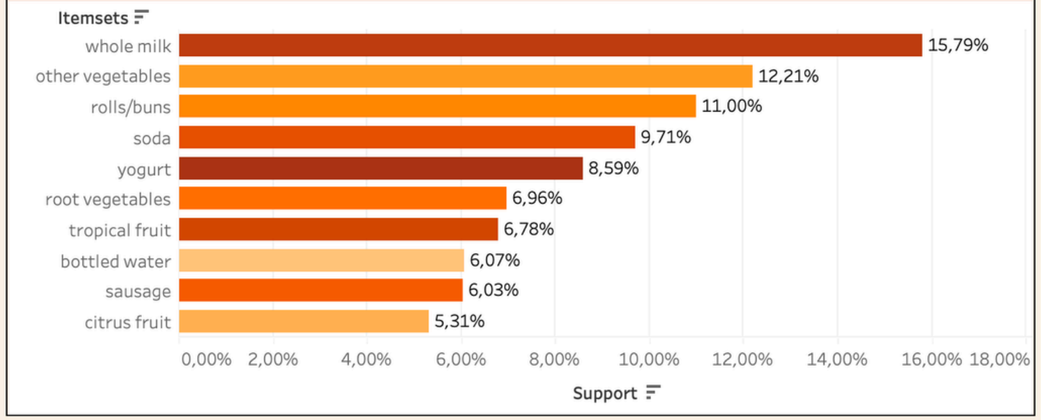
Market Basket Analysis

Total Transactions	Total Products	Item per Transaction	Rules Lift >1
14.963	167	2,59	24

Transaction Trend



Most bought products

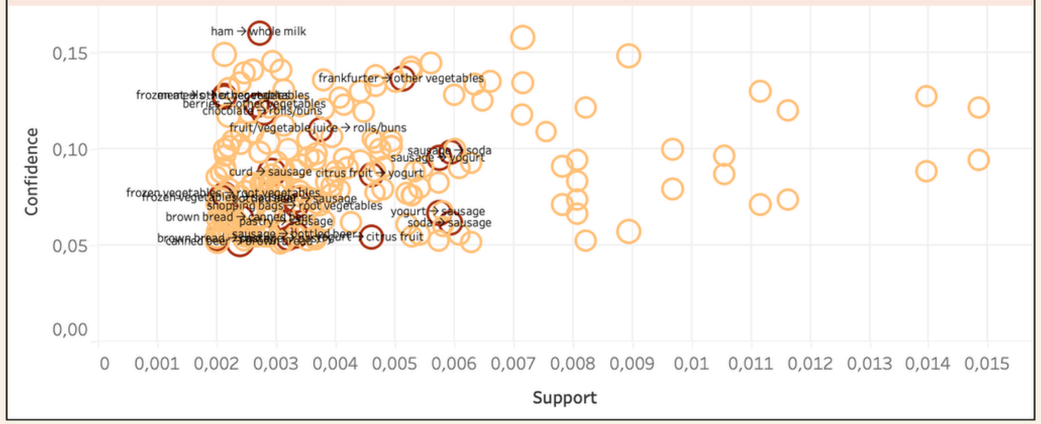


Products relations

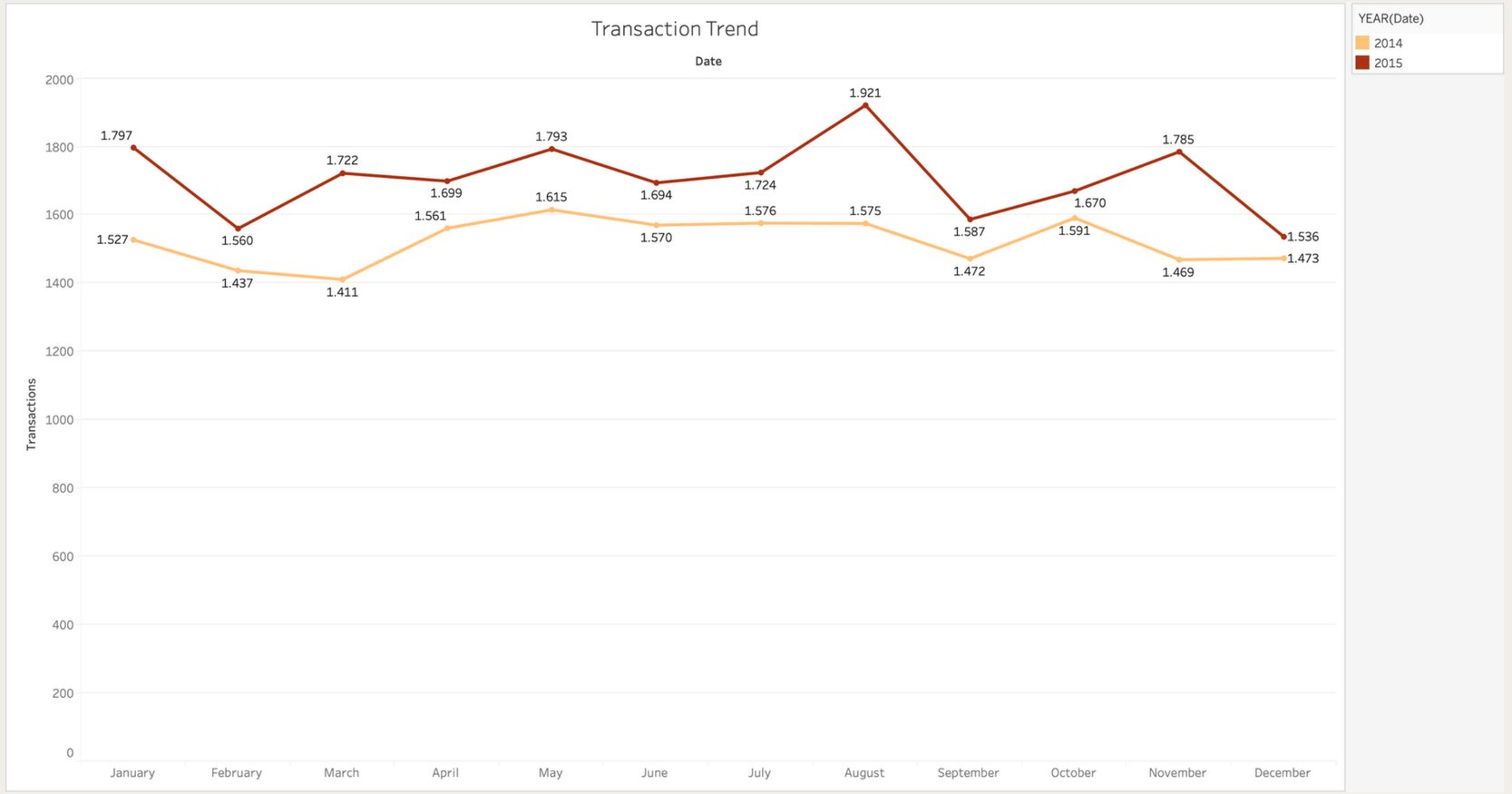
Consequents

Antecedents	bottled water	other vegetables	pastry	rolls/buns	root vegetables	sausage	soda	tropical fruit	whole milk	yogurt
bottled beer	0,92	0,85		0,80	0,78	1,22	0,67	0,83	1,00	0,88
bottled water		0,74		0,70	0,82	0,86	0,82	0,86	0,75	0,73
canned beer		0,70		0,82		0,87	0,69	0,80	0,81	0,96
domestic eggs	0,95	0,78		0,84	0,78		0,71	0,85	0,90	0,65
frankfurter	0,96	1,12		0,88	0,84		0,84		0,89	0,78
frozen vegetables		0,92		0,89	1,10	1,23	0,74		0,86	0,86
pastry	0,92	0,58		0,69	0,80	1,03	0,81	0,80	0,79	0,81
sausage	0,86	0,82	1,03	0,81	0,80		1,01	0,75	0,94	1,11
shopping bags		0,85		0,91	1,01		0,95	0,85	0,84	0,87
yogurt		0,77		0,83		1,11	0,70	0,90	0,82	

Rules that has Lift >1



DATA ANALYSIS OUTPUT



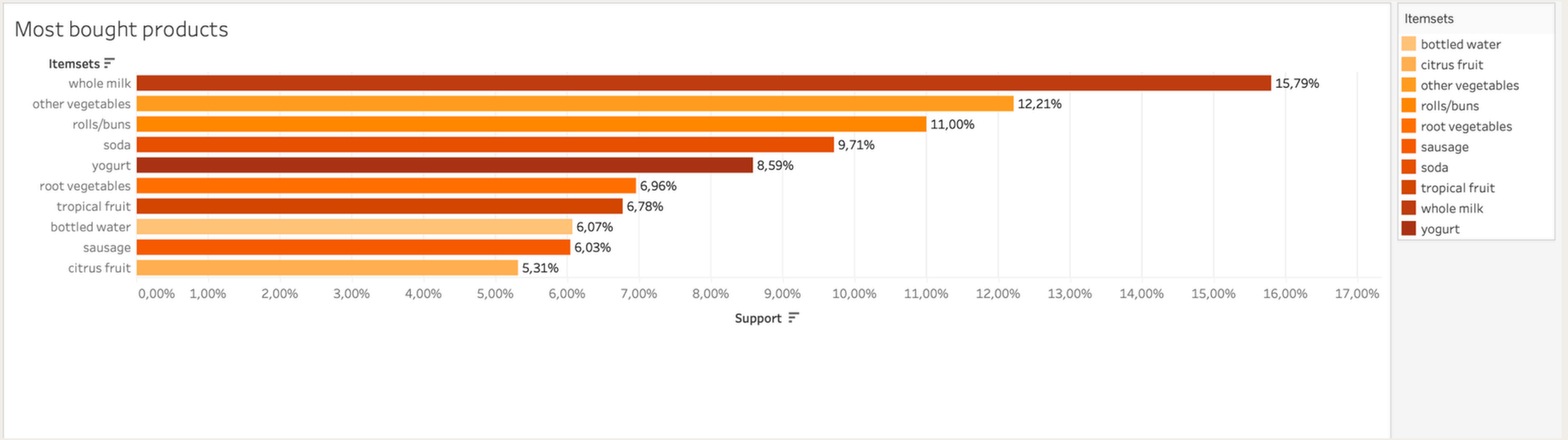
Transaction Growth

Year of Date	
2014	18.277
2015	20.488



Sepanjang 2015, toko menjual sekitar 12% lebih banyak barang dibanding 2014, dan kenaikan ini terjadi di semua bulan.

DATA ANALYSIS OUTPUT



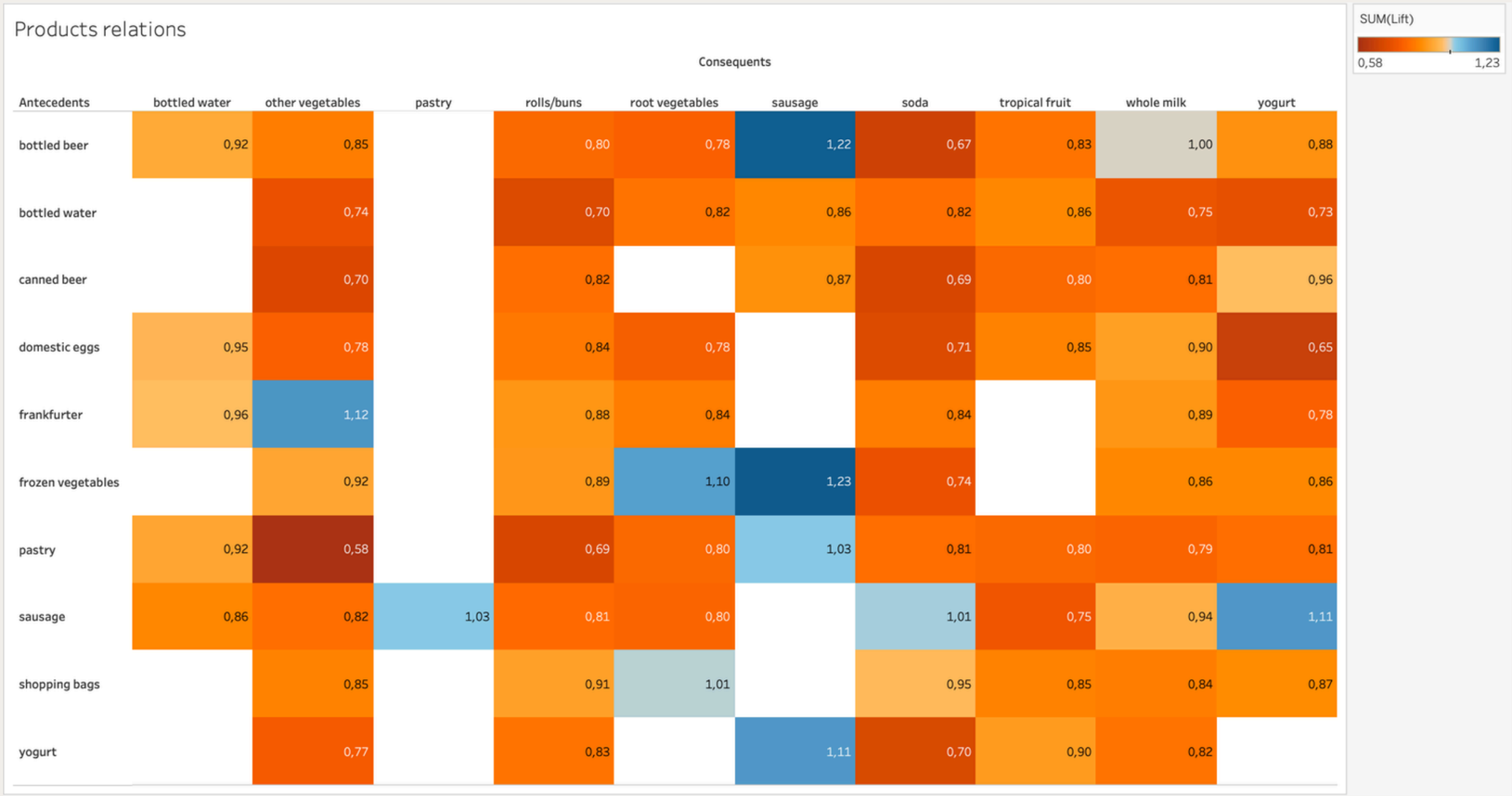
“Produk-produk ini seperti kios paling ramai di pusat jajan. Pertanyaan berikutnya: apakah orang yang mampir ke satu kios ramai biasanya juga mampir ke kios ramai lainnya?”

Berikut merupakan product yang paling banyak terjual

Untuk memahami chartnya lebih sederhana, bayangkan seratus pelanggan yang lewat. Sekitar 16 orang membeli susu, 12 orang membeli sayuran, dan 11 orang membeli roti. Inilah tiga kios paling ramai di toko ini. Sebaliknya, jeruk hanya dibeli sekitar 5 dari 100 orang. Jadi ada beberapa produk andalan yang paling sering masuk keranjang.



DATA ANALYSIS OUTPUT



Antecedents	bottled water	other vegetables	pastry	rolls/buns	root vegetables	sausage	soda	tropical fruit	whole milk	yogurt
bottled beer	0,92	0,85		0,80	0,78	1,22	0,67	0,83	1,00	0,88
bottled water		0,74		0,70	0,82	0,86	0,82	0,86	0,75	0,73
canned beer		0,70		0,82		0,87	0,69	0,80	0,81	0,96
domestic eggs	0,95	0,78		0,84	0,78		0,71	0,85	0,90	0,65
frankfurter	0,96	1,12		0,88	0,84		0,84		0,89	0,78
frozen vegetables		0,92		0,89	1,10	1,23	0,74		0,86	0,86
pastry	0,92	0,58		0,69	0,80	1,03	0,81	0,80	0,79	0,81
sausage	0,86	0,82	1,03	0,81	0,80		1,01	0,75	0,94	1,11
shopping bags		0,85		0,91	1,01		0,95	0,85	0,84	0,87
yogurt		0,77		0,83		1,11	0,70	0,90	0,82	

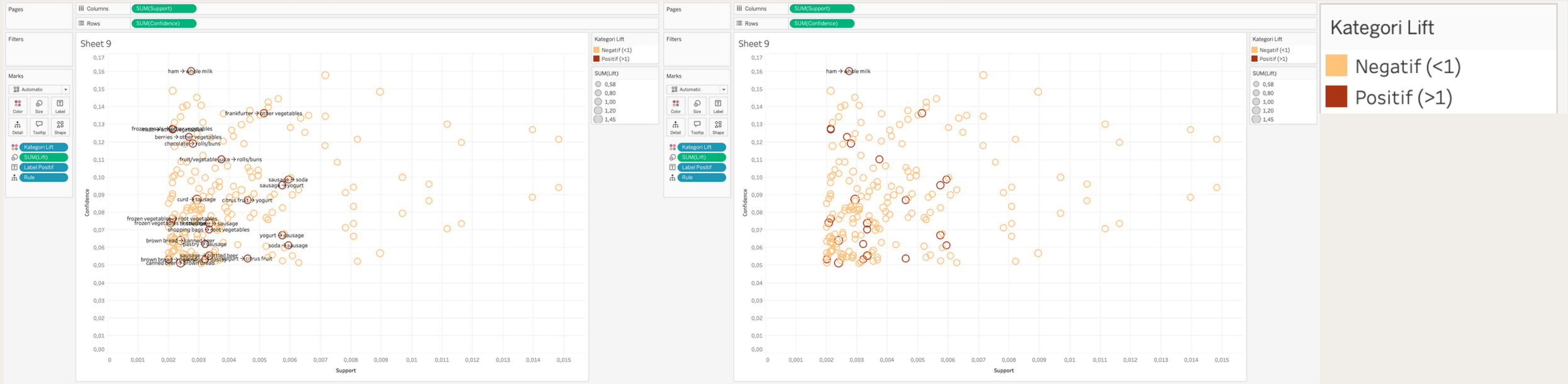
Bayangkan toko ini sebuah pesta, dan tiap produk adalah tamu. Heatmap ini menunjukkan siapa yang akrab dengan siapa. Warna oranye berarti dua tamu jarang berbincang, dan biru berarti mereka memang sering bersama. Hampir seluruh peta berwarna oranye, jadi kebanyakan produk dibeli sendiri-sendiri. Hanya segelintir pasangan yang biru, dan sausage menonjol sebagai tamu yang paling mudah bergaul: ia sering ditemani bir, soda, dan yogurt.

“kenapa 1,00 atau 1,03 tidak dianggap kuat?”

Itu seperti dua orang yang kebetulan duduk bersebelahan. Selisihnya terlalu tipis dari kebetulan, jadi belum bisa disebut janjian.” Fokus cerita sebaiknya pada angka yang jelas di atas 1, seperti 1,10 sampai 1,23



DATA ANALYSIS OUTPUT



Ringkasan chart

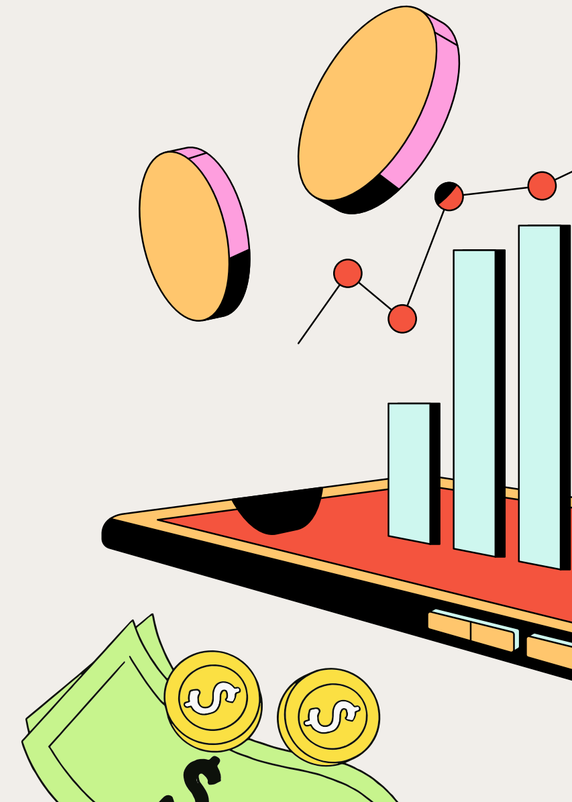
- Setiap titik = satu pasangan produk (misalnya curd dan sausage).
- Titik maroon = pasangan yang cenderung dibeli bersama (lift > 1).
- Titik orange muda = pasangan yang tidak istimewa (lift ≤ 1).
- Dari 219 titik, hanya 24 yang maroon.

“Bayangkan saya menebak isi keranjang pelanggan. Menebak ‘pasti beli susu’ sering benar, tapi hanya karena susu memang dibeli banyak orang, jadi itu bukan tebakan benar-benar tepat. Itulah titik-titik orange muda. Dari 219 tebakan, hanya 24 yang lebih jitu dari asal tebak, yaitu titik maroon”

Conclusion and Recommendations



1. Buat program promosi mingguan untuk produk kebutuhan sehari-hari, seperti susu, sayuran, dan roti, agar pelanggan memiliki alasan untuk kembali berbelanja secara rutin. Berikan voucher untuk kunjungan berikutnya yang berlaku dalam tujuh hari setelah pembelian. Strategi ini bertujuan mendorong pelanggan melakukan transaksi lebih sering, bukan hanya membeli lebih banyak produk dalam satu kunjungan.
2. Prioritaskan stok dan penempatan whole milk, other vegetables, dan rolls/buns karena ketiganya termasuk produk yang paling sering dibeli. Pastikan produk mudah ditemukan, rak tertata rapi, dan pengisian ulang dilakukan sebelum jam ramai.
3. Jangan langsung membundel semua produk terlaris karena hasil analisis menunjukkan sebagian besar pasangan produk tersebut tidak memiliki kecenderungan pembelian bersama yang kuat. Sebagai alternatif, uji promosi kombinasi berdasarkan kebutuhan pelanggan, serta jalankan promosi dalam periode terbatas dan bandingkan hasil penjualan produk pendamping sebelum dan selama promosi.



Thank You Very Much

+62 813 8545 8346

MUHAMMADALFI880@GMAIL.COM

HTTPS://WWW.LINKEDIN.COM/IN/ALFISYAHPUTRA

https://github.com/Alfi-Syahputra/Market-Basket-Analysis

